11.3 Areas of Polygons

Geometry 11.3 · Student sheet

Name _____ Period ____ Date _____

I can…

- find the area of a rhombus or a kite from its diagonals.
- find the central angle of a regular polygon.
- find the area of a regular polygon using its apothem and perimeter.

Vocabulary · fill in the missing word

Center of a regular polygon — the center of its _____ circle

Radius of a regular polygon — the radius of its _____ circle

Apothem of a regular polygon — the _____ from the center to any side of the polygon

Central angle of a regular polygon — an angle formed by two radii drawn to _____ vertices

Part 1 · Discovery: a hexagon is six triangles

Draw a circle with a 5 cm radius, step the compass around it to make a regular hexagon, and draw the six radii and one apothem.

Side	Apothem	Area of one triangle	Area of the hexagon (6 triangles)	½ · apothem · perimeter

What do you notice about the last two columns?

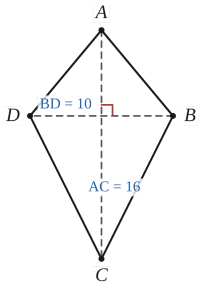
Part 2 · Notes and examples (fill in each note, then work the example)

Area of a rhombus or kite: A = ½ _____, where d_1 and d_2 are the diagonals.

Example 1

a. Find the area of the kite.

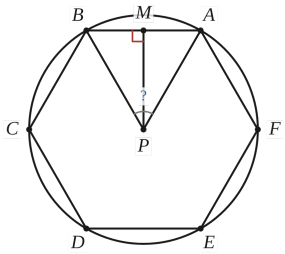
b. A rhombus has diagonals of 6 and 9. Find its area.



Central angle of a regular n-gon: _____

Example 2

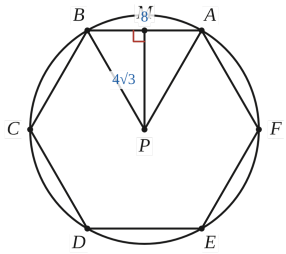
In regular hexagon ABCDEF with center P, M is the midpoint of $\overline{AB}$. Find m∠APB and m∠APM.



Area of a regular polygon: A = ½ · a · P, where a is the _____ and P is the _____.

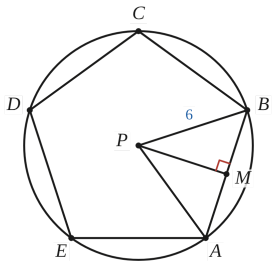
Example 3

A regular hexagon has sides of 8 inches and an apothem of $4\sqrt{3}$ inches. Find its area. Round to the nearest hundredth.



Example 4

A regular pentagon has a radius of 6. Find its apothem, side length, and area. Round to the nearest tenth.



Example 5

A stop sign is a regular octagon with sides of 10 inches and an apothem of about 12.07 inches. Find its area. Round to the nearest tenth.

Part 3 · Partner practice

Partner 1

A rhombus has diagonals of 12 and 7. Find its area.

Partner 2

A regular hexagon has sides of 6 and an apothem of $3\sqrt{3}$. Find its area. Round to the nearest hundredth.